DNS Records and Messages

DNS Resource Records (RRs)

A **Resource Record (RR)** is the basic unit of information in DNS. Each RR is a **4-tuple**:

$$(Name, Value, Type, TTL)$$

- **Name** – Domain or hostname.
- **Value** – Data depending on record type.
- **Type** – Specifies kind of record (A, NS, MX, etc.).
- **TTL (Time-to-Live)** – Lifetime of record in cache.

Common DNS Record Types

Essential Record Types

- **A** – Maps hostname → IPv4 address. Example: `(relay1.bar.foo.com, 145.37.93.126, A)`
- **AAAA** – Maps hostname → IPv6 address. Example: `(example.com, 2001:db8::1, AAAA)`
- **NS** – Delegates a domain to an authoritative DNS server. Example: `(foo.com, dns.foo.com, NS)`
- **CNAME** – Maps alias → canonical hostname. Example: `(foo.com, relay1.bar.foo.com, CNAME)`
- **MX** – Mail Exchange; maps domain → mail server hostname. Example: `(foo.com, mail.bar.foo.com, MX)`
- **PTR** – Pointer record (reverse DNS lookup: IP → domain). Example: `(126.93.37.145.in-addr.arpa, relay1.bar.foo.com, PTR)`
- **SOA** – Start of Authority; defines the authoritative information about a DNS zone (primary server, admin contact, serial number, timers).
- **TXT** – Stores human-readable or machine-readable text. Often used for SPF, DKIM, or verification codes.
- **SRV** – Service record; defines hostname/port for specific services. Example: `(_sip._tcp.example.com, sipserver.example.com:5060, SRV)`
- **NAPTR** – Naming Authority Pointer; used for more advanced service discovery (e.g., VoIP, ENUM).

Authority and Caching

- If a DNS server is **authoritative**, it contains a Type A record for the hostname.
- If not authoritative, it stores:
  - An **NS record** pointing to the authoritative server.
  - An **A record** mapping the DNS server's hostname → IP address.

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DNS Messages

DNS uses only **two kinds of messages**:

1. **Query Message** – Sent by client.
2. **Reply Message** – Sent by server.

Both have the same format (see Figure 1).

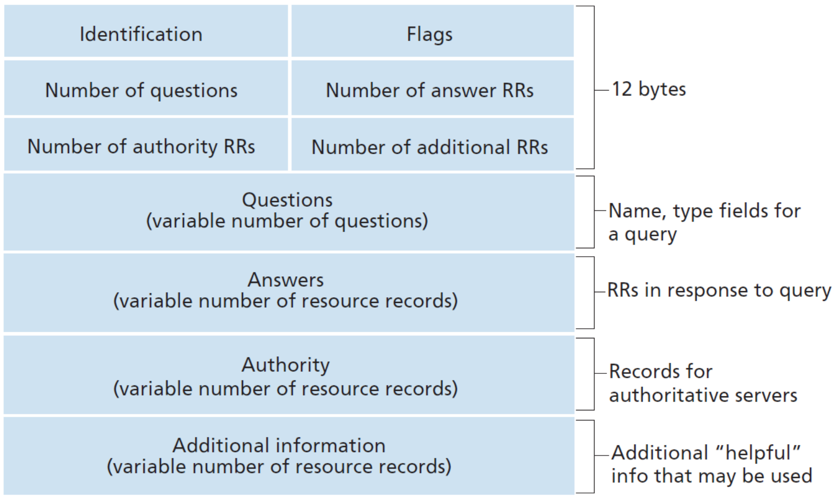

Figure 1: DNS Message Format (simplified)

Message Structure

Header (12 bytes)

- **Identification (16 bits)** – Match query ↔ reply.
- **Flags (16 bits)**:
  - QR (1 bit): 0=query, 1=reply.
  - Opcode (4 bits): Query type (standard, inverse, status).
  - AA (1 bit): Authoritative Answer.

- TC (1 bit): Truncation (message too long).
- RD (1 bit): Recursion Desired.
- RA (1 bit): Recursion Available.
- Z (3 bits): Reserved (set to 0).
- RCODE (4 bits): Response code (e.g., no error, name error).

- **QDCOUNT (16 bits)** – Number of questions.
- **ANCOUNT (16 bits)** – Number of answer RRs.
- **NSCOUNT (16 bits)** – Number of authority RRs.
- **ARCOUNT (16 bits)** – Number of additional RRs.

Question Section

- **QNAME** – Domain name being queried.
- **QTYPE** – Type of query (A, MX, etc.).
- **QCLASS** – Typically IN (Internet).

Answer Section

- Contains RRs answering the query.
- Each RR has: `Name, Type, Class, TTL, Data length, RDATA`.

Authority Section

- Provides NS records of authoritative servers for the queried domain.

Additional Section

- Provides helpful extra records (e.g., A/AAAA records for NS or MX servers).

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Inserting Records into DNS Database

When registering a new domain, the following process ensures the domain becomes resolvable worldwide.

Step-by-Step Process

1. **Domain Registration:** A domain (e.g., `networkutopia.com`) is registered through a **Registrar** accredited by ICANN.
2. **Authoritative Servers Specification:** The registrant provides names and IPs of primary and secondary **authoritative DNS servers**.
3. **Registrar's Role:** Registrar communicates with the **TLD servers** (e.g., `.com`) and inserts:
   - **NS records** for the domain.
   - **A records** mapping authoritative DNS server names to their IPs.
4. **Configuring Authoritative Servers:** The domain owner adds actual resource records:
   - **A/AAAA records** for web servers.
   - **MX records** for mail servers.
   - **CNAME records** for aliases.
   - **TXT records** for SPF/DKIM or verification.
5. **Propagation:** Records spread across recursive resolvers worldwide according to the **TTL**.

Illustrative Example

Example: NetworkUtopia Startup

Domain: `networkutopia.com`
Authoritative Servers:

- `dns1.networkutopia.com` → `212.212.212.1`
- `dns2.networkutopia.com` → `212.212.212.2`

Registrar Inserts into TLD:

- `(networkutopia.com, dns1.networkutopia.com, NS)`
- `(dns1.networkutopia.com, 212.212.212.1, A)`

Authoritative Servers Store:

- `(www.networkutopia.com, 212.212.71.4, A)`
- `(mail.networkutopia.com, 212.212.71.5, MX)`
- `(ftp.networkutopia.com, www.networkutopia.com, CNAME)`

Key Insight

The registrar only stores NS and A records at the TLD level. All other records must be maintained in the authoritative servers.

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DNS Vulnerabilities

DNS is a critical part of the Internet infrastructure but is also a frequent target for attacks.

Types of Attacks

Attack Type	Description & Example
DDoS Attacks	Flood DNS servers with queries. *Example:* 2002 Root Server attack, 2016 Dyn Mirai botnet attack.
Amplification Attacks	Spoof victim's IP in small queries $\rightarrow$ servers send large responses to victim. DNS used as a reflector to overwhelm target.
Cache Poisoning	Inserts bogus records into resolver cache. *Example:* Kaminsky attack (2008).
MITM (Man-in-the-Middle)	Intercepts queries and forges fake responses (e.g., fake bank IP redirection).

Table 1: Major DNS vulnerabilities

Mitigation Techniques

- **DNSSEC (DNS Security Extensions):** Adds digital signatures ensuring authenticity and integrity.
- **Rate Limiting:** Prevents amplification by limiting query rates.
- **Anycast Routing:** Distributes DNS servers globally to absorb DDoS traffic.

Important Note

DNSSEC prevents forgery and cache poisoning but does not provide confidentiality — queries are still visible.

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Verification Example: DNS Resolution Walkthrough

Suppose Alice (in Australia) queries `www.networkutopia.com`.

Step-by-Step Resolution

1. **Local DNS Query:** Alice's computer asks her local recursive resolver.
2. **Root Server Referral:** Resolver queries a root server, which points to the `.com` TLD servers.
3. **TLD Server Reply:** Returns:
   - `(networkutopia.com, dns1.networkutopia.com, NS)`
   - `(dns1.networkutopia.com, 212.212.212.1, A)`
4. **Authoritative Query:** Resolver contacts `dns1.networkutopia.com`.
5. **Authoritative Response:** Returns: `(www.networkutopia.com, 212.212.71.4, A)`.
6. **Client Connection:** Alice's browser connects via TCP to `212.212.71.4`.

Visual Flow

DNS Resolution Flow

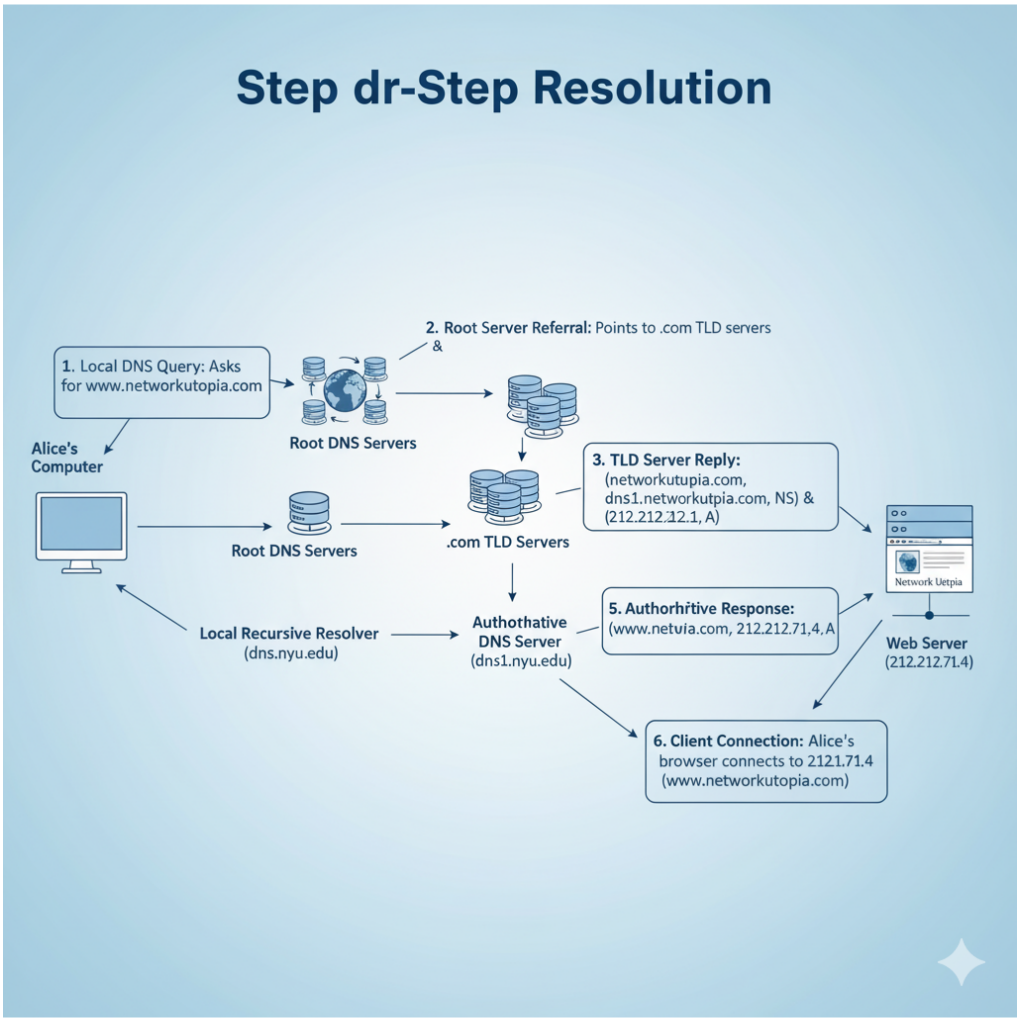

Figure 2: DNS Resolution Path for `www.networkutopia.com`

Key Takeaway

DNS resolution follows a hierarchy: **Root** → **TLD** → **Authoritative** → **Client**.

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Key Takeaways

Summary

- **Domain registration** requires NS and A records at TLD servers and configuration of authoritative DNS servers.
- **DNS attacks** include DDoS, cache poisoning, and MITM exploits.
- **DNSSEC** provides integrity and authenticity through digital signatures but not confidentiality.
- DNS resolution follows a hierarchical path: Root → TLD → Authoritative → End-user.